

Nikhil Sobanbabu

Robotics Research | Reinforcement Learning | Sim-to-Real Control

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Robotics researcher focused on learning-based control, system identification, and sim-to-real deployment for humanoids and quadrupeds. Co-first-author publications at CoRL and ICAPS.

RESEARCH AND INDUSTRY EXPERIENCE

Amazon FAR

MTS Intern

May 2026 – Present

- Developing cross-trial behavior adaptation for humanoid loco-manipulation tasks.
- Integrating Fauna Sprout with Holosoma.

Carnegie Mellon University — LeCAR Lab

Graduate Research Assistant; Advisor: Prof. Guanya Shi

May 2024 – May 2026

Pittsburgh, PA

- Co-developed **FADA**, a planner-inverse dynamics framework for few-shot humanoid adaptation; fine-tunes the inverse dynamics model using approximately two minutes of target-domain rollouts. Co-first author; CoRL 2026 (accepted).
- Developed **SPI-Active**, a parallel sampling-based system identification framework that estimates inertial and actuator parameters by minimizing open-loop trajectory prediction error.
- Designed Fisher-information-based active exploration for informative data collection; enabled sim-to-real locomotion policies that outperformed baselines by **42–63%**. Co-first author; CoRL 2025 Oral.

Carnegie Mellon University — Search-Based Planning Lab

Graduate Research Assistant; Advisor: Prof. Maxim Likhachev

Jan 2025 – May 2026

Pittsburgh, PA

- Developed waypoint refinement and synchronization with custom locomotion policies for coordinated navigation of heterogeneous quadrupeds using Conflict-Based Search and Temporal Plan Graphs.
- Built a modular Unitree sim-to-real control stack supporting multi-agent deployment and switching between high- and low-level controllers; co-first author of an ICAPS 2025 demo.

University of Southern California — DRCL

Research Intern; Advisor: Prof. Quan Nguyen

May 2023 – May 2026

Los Angeles, CA

- Developed a closed-loop reference generator with preview control and centroidal dynamics for reinforcement learning of HECTOR biped parkour in MuJoCo.
- Extended oracle-guided policies with hybrid automata for whole-body soccer and box manipulation across HECTOR, Berkeley Humanoid, and Unitree G1; co-author of IROS 2025 work.

EDUCATION

Carnegie Mellon University

M.S., Electrical and Computer Engineering; GPA: 4.0/4.0

Aug 2024 – May 2026

Pittsburgh, PA

Indian Institute of Technology Madras

B.Tech., Electrical Engineering; GPA: 9.39/10.0

Aug 2020 – May 2024

Chennai, India

HONORS AND SERVICE

K. C. Mahindra Graduate Scholarship (2024); IUSSTF–Viterbi Research Internship (2023); JEE Advanced All India Rank 286 (2020). Reviewer: ICRA, IROS, RAL, CoRL. Teaching assistant in feedback control, mathematical foundations, and control engineering.

ADDITIONAL EXPERIENCE

DiFACTO Robotics and Automation

Robotics Intern

Jun 2024 – Aug 2024

Bengaluru, India

- Built natural-feature navigation and recovery for in-house autonomous mobile robots using ROS Navigation Stack.
- Optimized FANUC R-1000iA/80H palletization trajectories, reducing cycle time by **23.6%**.

IIT Madras — Control Engineering Laboratory

Undergraduate Research Assistant; Advisor: Prof. Bharath Bhikkaji

Aug 2022 – May 2024

Chennai, India

- Derived reach-avoid strategies using Hamilton–Jacobi–Isaacs PDEs, validated three-agent experiments on LEGO EV3, and compared MAPPO learning with analytical Nash equilibria.

SELECTED PUBLICATIONS

* Equal contribution. Full publication list: nike353.github.io.

FADA: Few-Shot Domain Adaptation via Dynamics Alignment for Humanoid Control

Angchen Xie*, **Nikhil Sobanbabu***, Ishayu Shikhare, Alan Wang, Max Simchowitz, Guanya Shi[CoRL 2026 \(accepted\)](#) | [Paper](#)

Sampling-Based System Identification with Active Exploration for Legged Robot Sim2Real Learning

Nikhil Sobanbabu*, Guanqi He*, Tairan He, Yuxiang Yang, Guanya Shi[CoRL 2025 \(Oral\)](#) | [Paper](#)

Towards Unstructured MAPF: Multi-Quadruped MAPF Demo

Rishi Veerapaneni*, **Nikhil Sobanbabu***, Guanya Shi, Jiaoyang Li, Maxim Likhachev[ICAPS 2025 \(Demo Track\)](#) | [Paper](#)

Additional co-authored work: RIO (RSS 2026); ASAP (RSS 2025); Preferred OGMP (IROS 2025); HDMI (2025 preprint); OGMP (2024 preprint).

SELECTED PROJECTS

Adaptive force control for quadrupeds

Spring 2025

Integrated L1-adaptive control with hierarchical RL and QP control; achieved **2.5–3×** longer continuous-jumping traversal under sim-to-sim payload changes. Code

Motion imitation with DIAL-MPC

Fall 2024

Retargeted dog motion capture to Unitree Go2 and implemented sampling-based MPC, improving trajectory-tracking performance by **22.73%**.

LEADERSHIP

Team Anveshak, IIT Madras — Team Lead (2022–2023); member (2021–2023). Led **40+ students** to **6th globally** at the 2022 Anatolian Rover Challenge; developed CAN-based rover electronics and power monitoring.

TECHNICAL SKILLS

Programming: Python, C/C++, Julia, Bash, Git.

Learning and optimization: PyTorch, TensorFlow, NumPy, scikit-learn, CasADi.

Robotics and simulation: Isaac Gym/Sim, MuJoCo, ROS 1/2, Gazebo, MoveIt, MATLAB/Simulink.

Systems: Docker; STM32, Arduino, NVIDIA Jetson; Altium PCB design; sensor and actuator integration.